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1. Prove Fermat's Little Theorem and explain Miller-Rabin. Apply it to 561 and explain why 561 fools Fermat test.

Ans: to the Question No - 01

Fermat's Little Theorem:

Let p be a prime and a be an integer

such that $p \nmid a$

Consider the non-zero elements:-

$1, 2, 3, \dots, p-1$

Modulo p , multiplication by a only rearranges these elements because a has an inverse modulo p . Therefore

$a \cdot 1, a \cdot 2, \dots, a \cdot (p-1)$

are the same elements as

$1, 2, \dots, p-1$

in some order.

so,

$$a^{p-1} (p-1)! \equiv (p-1)! \pmod{p}$$

Since p is prime $(p-1)!$ has an inverse modulo p . Therefore,

$$a^{p-1} \equiv 1 \pmod{p}$$

This is Fermat's Little theorem.

Relationship with Miller - Rabin:

Fermat's theorem says that for a prime p , $a^{p-1} \equiv 1 \pmod{p}$

so, we can use this idea to test whether a number is prime.

But some composite numbers also satisfy Fermat's condition.

Therefore - Miller - Rabin uses a stronger test by writing,

$$n-1 = 2^s d$$

and checking whether,

$$ad \equiv 1 \pmod{n} \text{ or}$$

$$a^{2^s d} \equiv -1 \pmod{n}$$

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$$n-1 = 2^s d$$

and checking whether,

$$a^d \equiv 1 \pmod{n} \text{ or}$$

$$a^{2^r d} \equiv -1 \pmod{n}$$

for some $0 \leq r \leq s$
Miller-Rabin for $n = 561$ has $a = 2$

$$561 - 1 = 560 = 2^4 \times 35$$

$$\text{So, } s = 4, d = 35$$

$$\text{calculate: } 2^{35} \pmod{561} = 263$$

This Repeatedly square

$$2^{70} \equiv 166 \pmod{561}$$

$$2^{140} \equiv 67 \pmod{561}$$

$$2^{280} \equiv 1 \pmod{561}$$

For miller-Rabin, we need either
1 initially or $-1 \equiv 560$ at one of
the required stages. Here we get

$$263, 166, 67, 1$$

There is no -1 .

Therefore 561 is composite.

561 satisfies $a \not\equiv 0 \pmod{561}$

$$2^{560} \equiv 1 \pmod{561}$$

$$561 = 3 \times 11 \times 17$$

$$a^{n-1} \equiv 1 \pmod{n}$$

for every a relatively prime to n .

2. Prove $\mathbb{Z}_n$ is a field iff n is prime
compute $\mathbb{Z}_{11}$ and $\mathbb{Z}_{12}$.

Ans. to the Question No-02

We need to prove both direction.
iff n is prime.

Let $n = p$, where p is prime.

The elements are,

$$\mathbb{Z}_p = \{0, 1, 2, \dots, p-1\}$$

For every non-zero element a , we have
 $\gcd(a, p) = 1$

Therefore, a has a multiplicative inverse modulo p .

Hence every non zero element has an inverse. So,

$\mathbb{Z}_p$ is a field.

if $\mathbb{Z}_n$ is a field.

Suppose n is composite

Then $n = ab$ where, $1 < a < n$
and $1 < b < n$

Therefore, $a \cdot b \equiv 0 \pmod{n}$

although $a \neq 0$ and $b \neq 0$

These are called zero divisors.

A field cannot contain zero divisors.

Therefore n cannot be composite

Hence, $\mathbb{Z}_n$ is a field iff n is prime

Comparison:

$\mathbb{Z}_{11}$	$\mathbb{Z}_{12}$
1) 11 is prime	1) 12 is composite
2) It is a field	2) It is not a field
3) every non-zero element has inverse	3) Some element has no inverse
4) No zero divisors	4) Has zero - divisors
5) Example: $3^{-1} = 4$ because $3 \times 4 = 12 \equiv 1$	5) Example $3 \times 4 = 12 \equiv 0$

Therefore $\mathbb{Z}_{11}$ is a field but

$\mathbb{Z}_{12}$ is not field

3. Construct $\text{GF}(2^3)$ using x^3+x+1 , find a generator and show the multiplicative group is cyclic.

Ans: to the Question No - 03

Note: The question says $\text{GF}(2^3)$, but means $\text{GF}(2^3) = \text{GF}(8)$.

A Let $f(x) = x^3 + x + 1$

over $\text{GF}(2)$.

Since $f(x)$ is irreducible over $\text{GF}(2)$

We can construct,

$$\text{GF}(2^3) = \text{GF}(8)$$

Let α be a root of $f(x)$.

Therefore, $\alpha^3 + \alpha + 1 = 0$

In $\text{GF}(2)$

$$-1 = 1 \text{ so}$$

$$\alpha^3 = \alpha + 1$$

The elements are:

$$\text{GF}(8) = \{0, 1, \alpha, \alpha^2, \alpha^3, \alpha^4, \alpha^5, \alpha^6\}$$

using $\alpha^3 = \alpha + 1$

We get: $\alpha^4 = \alpha^2 + \alpha$

$$\alpha^5 = \alpha^2 + \alpha + 1$$

$$\alpha^6 = \alpha^2 + 1$$

and $\alpha^7 = 1$

Therefore α has order 7.

since $\alpha^i \alpha^j = \alpha^{i+j}$

and exponents are taken modulo 7:

α	1	α	α^2	α^3	α^4	α^5	α^6
1	1	α	α^2	α^3	α^4	α^5	α^6
α	α	α^2	α^3	α^4	α^5	α^6	1
α^2	α^2	α^3	α^4	α^5	α^6	1	α
α^3	α^3	α^4	α^5	α^6	1	α	α^2
α^4	α^4	α^5	α^6	1	α	α^2	α^3
α^5	α^5	α^6	1	α	α^2	α^3	α^4
α^6	α^6	1	α	α^2	α^3	α^4	α^5

There are $\phi(8) = 4$ non zero elements
 since $1, 3, 5, 7$ and
 no smaller positive power of α gives 1

Therefore α generates all non-zero elements.
 $\text{order}(\alpha) = 4$

$\text{GF}(8)^*$ is cyclic

4. Find the complexity of modular
 exponentiation and calculate 7580
 $\text{mod } 561$

Ans: to the Question No-03

Square and Multiply

$a^e \pmod n$

We write e in binary

$e = 561 = 1000101001$

$1000101001 = 2^{10} + 2^9 + 2^7 + 2^5 + 2^3 + 2^1 + 1$

Binary Representation is as follows

$$560 = (1000110000)_2$$

The algorithm uses:

- 1) Squaring for every binary bit
- 2) Multiplication when the bit is 1

The number of operations is proportional to the number of bits of a .

Therefore, $O(\log a)$

time complexity

$$\text{calculate } 7^{56} \pmod{561}$$

$$7^2 = 49$$

$$7^4 = 49^2 = 2401 \equiv 157 \pmod{561}$$

$$7^8 = 157^2 \equiv 526 \pmod{561}$$

$$7^{16} \equiv 526^2 \equiv 103 \pmod{561}$$

$$7^{32} \equiv 103^2 \equiv 511 \pmod{561}$$

$$7^{64} \equiv 511^2 \equiv 1257 \pmod{561}$$

$$7^{128} \equiv 256^2 \equiv 460 \pmod{561}$$

$$7^{256} \equiv 460^2 \equiv 103 \pmod{561}$$

Since:

$$561 = 256 + 128 + 64 + 64 + 32 + 16$$

or more simply, because 561 is a Carmichael number and

$$\gcd(7, 561) = 1 \text{ we have}$$

$$7^{560} \equiv 1 \pmod{561}$$

Interpretation:

561 is composite:

$$561 = 3 \times 11 \times 17 \text{ but still,}$$

$$7^{560} \equiv 1 \pmod{561}$$

This shows why the simple Fermat primality test can fail for Carmichael numbers.

5. Compare AKS and Miller-Rabin with RSA

Feature	AKS	Miller-Rabin
Type	Deterministic	Probabilistic
Result	Always correct	Very small probability of error
Main idea	Polynomial identity	Modular exponentiation
Error	No error	Error can be reduced by more rounds
RSA use	Rare in practice	commonly used
Speed	Theoretically polynomial but expensive	very fast in practice

Complexity

AKS has polynomial time complexity in the number of bits of n . Its theoretical complexity has been improved over time, but it is still generally much less practical than Miller-Rabin for ordinary RSA key generation.

Miller-Rabin mainly uses modular exponentiation, with approximately,

$$O(k \log^3 n)$$

$$O(1) \cdot 2^0 = 0.10$$

$$2 + (1) \cdot 0.1 = 0.12$$

$$1 + (1) \cdot 0.12 = 0.13$$

$$5 + (1) \cdot 0.13 = 0.14$$

$$0 + (1) \cdot 0.14 = 0.15$$

$$5 = (0.15 \cdot 10) \log$$

6. Derive Extended Euclidean Algorithm and find $\text{gcd}(240, 46)$ with Bezout coefficient x, y to express it as

Ans. to the question No-06

The Euclidean Algorithm based on

$a = bq + r$
where, q is the quotient and r is the Remainder.

For extended Euclidean Algorithm, We express every remainder as a linear combination of a and b .

Step: Euclidean Algorithm :

$$240 = 46(5) + 10$$

$$46 = 10(4) + 6$$

$$10 = 6(1) + 4$$

$$6 = 4(1) + 2$$

$$4 = 2(2) + 0$$

Therefore, $\text{gcd}(240, 46) = 2$

Step 2: Back substitution

From $6 = 4 + 2$

We get, $2 = 6 - 4$

But $4 = 10 - 6$

Therefore, $2 = 6 - (10 - 6)$

$$2 = 2(6) - 10$$

Now,

$$6 = 46 - 4(10)$$

So,

$$2 = 2(46 - 4(10)) - 10$$

$$2 = 2(46) - 9(10) \text{ and}$$

$$10 = 240 - 5(46)$$

Therefore,

$$2 = 2(46) - 9(240 - 5(46))$$

$$2 = 2(46) - 9(240) + 45(46)$$

$$2 = -9(240) + 47(46) \text{ Thus}$$

$$2 = 240(-9) + 46(47)$$

Comparing with,

$$240x + 46y = \gcd(240, 46)$$

We get, $x = -9, y = 47$

Final Ans: $\gcd(240, 46) = 2$

$$01 - (x = -9, y = 47)$$

$$(01)p - 2p = 2$$

7. State and prove CRT, then solve the four congruences, we have

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 4 \pmod{7}$$

$$x \equiv 5 \pmod{11}$$

$$(fp)2p + (e-1)0p = 2$$

Ans: to the Question No - 07

statement of CRT:

If $m_1, m_2, \dots, m_k$ are pairwise coprime, then the system

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

$$x \equiv a_k \pmod{m_k}$$

has:

- ① At least one solution and
- ② The solution is unique modulo

$$M = m_1 m_2 \dots m_k$$

Pro of: $\therefore$

Let $M = m_1 m_2 \dots m_k$ and

define $M_i = \frac{M}{m_i}$

Since the moduli are pairwise coprime,

$$\gcd(m_i, m_j) = 1$$

Therefore, m_i has a multiplicative inverse modulo m_j . Let

$$m_i y_i \equiv 1 \pmod{m_j}$$

Then construct:

$$x = \sum_{i=1}^k a_i m_i y_i$$

For a particular modulus m_j :

$$m_i y_i \equiv 1 \pmod{m_j}$$

All other m_i 's contain m_j , so they are $0 \pmod{m_j}$.

$$\text{Thus } x \equiv a_j \pmod{m_j}$$

So, a solution exists.

Uniqueness; $(11 \text{ } 1000) \div 2 = 5500$

Suppose x and y are two solutions.
Then for every i ,

$$x \equiv y \pmod{m_i}$$

Therefore, $m_i \mid (x-y)$

Since the m_i is pairwise coprime

$$M = m_1 m_2 \cdots m_k \mid (x-y)$$

Hence, $x \equiv y \pmod{M}$

Therefore, the solution is unique
modulo M .

[CRT is proved]

Applying CRT:

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 4 \pmod{7}$$

$$x \equiv 5 \pmod{11}$$

Step 1: Find M_i for each x_i

$$M = 3 \times 5 \times 7 \times 11$$

$$M = 1155$$

$$\text{Now, } M_1 = \frac{1155}{3} = 385$$

$$M_2 = \frac{1155}{5} = 231$$

$$M_3 = \frac{1155}{7} = 165$$

$$M_4 = \frac{1155}{11} = 105$$

Step 2: Find inverse

For $M_1 = 385$:

$$385 \equiv 1 \pmod{3}$$

$$\text{So, } y_1 = 1$$

For $M_2 = 231$:

$$231 \equiv 4 \pmod{5}$$

$$\text{So, } y_2 = 1$$

For $M_3 = 165$:

$$165 \equiv 4 \pmod{7}$$

We need: $4y_3 \equiv 1 \pmod{7}$

Since, $4 \times 2 \equiv 8 \equiv 1 \pmod{7}$

$$y_3 = 2$$

For $M_4 = 105$:

$$105 \equiv 6 \pmod{11}$$

We need $6y_4 \equiv 1 \pmod{11}$

$$6 \times 2 \equiv 12 \equiv 1 \pmod{11}$$

$$y_4 = 2$$

Step 3: Calculate x

CRT Formula,

$$x = a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3 + a_4 M_4 y_4$$

$$x = 2(385)(1) + 3(231)(1) + 4(165)(2) + 5(105)(2)$$

$$x = 770 + 693 + 1320 + 1050$$

$$x = 3833$$

Now reduce modulo 1155:

$$3833 - 3(1155) = 3833 - 3465$$

$$= 368$$

Therefore,

$$x = 368 \pmod{1155}$$

Verification:

$$368 \div 3 \Rightarrow \text{rem } 2$$

$$368 \div 5 \Rightarrow \text{rem } 3$$

$$368 \div 7 \Rightarrow \text{rem } 4$$

$$368 \div 11 \Rightarrow \text{rem } 5$$

so, all conditions are satisfied

$$\therefore x \equiv 9368 \pmod{1155}$$

8. CRT based RSA Decryption/ signing optimization -

Ans: to the Question No-08

In normal RSA, decryption/ signing is

$$m = c^d \pmod{n}$$

where, $n = pq$

CRT makes this faster by doing two smaller calculation.

step 1: calculate d_p and d_q

$$d_p = d \pmod{p-1}$$

$$d_q = d \pmod{q-1}$$

Step: $q \text{ Inv}$

$$(q \text{ Inv}) \equiv q^{-1} \pmod{p}$$

$$\text{So, } q \times q \text{ Inv} \equiv 1 \pmod{p}$$

Example:

$$\text{Let } p = 11, q = 17$$

$$\text{Then: } n = 11 \times 17 = 187$$

Suppose private exponent:

$$d \equiv 107$$

calculate,

$$dp = 107 \pmod{10} = 7$$

$$dq = 107 \pmod{11} = 11$$

Now calculate $q \text{ Inv}$

$$17^{-1} \pmod{11} = 7$$

$$\text{Since } 17 \equiv 6 \pmod{11}$$

$$\text{and } 6 \times 2 = 12 \equiv 1 \pmod{11}$$

$$q \text{ Inv} = 2$$

CRT de encryption :

suppose ciphertext,

$$c = 100$$

First calculate,

$$m_1 = c \bmod p = 100 \bmod 11$$

$$\text{since } 100 \equiv 1 \pmod{11}$$

Next,

$$m_2 = c \bmod q = 100 \bmod 17$$

$$\text{since } 100 \equiv 15 \pmod{17}$$

$$m_2 = 15$$

$$\text{Now, } h = \text{gcd}(m_1, m_2)$$

$$h = 2(1+15) \bmod 11$$

$$h = -28 \bmod 11 = 5$$

Finally, $m = m_2 + kq$

$$= 15 + (5)(17)$$

$$= 100$$

So, the decrypted message is $m = 100$

Why is it about 4x faster?

Without CRT, we calculate one exponential modulo $n = pq$.
With CRT, we calculate two - exponentiations.

1) Modulo p

2) Modulo q

Since p and q are about half the size of n , modular exponentiation with each is much cheaper.

9. Bellcore fault-Injection Attack on RSA - CRT

Ans: to the Question No-09

The Bellcore attack is a fault attack against RSA - CRT.

Normally RSA - CRT calculates:

$$S_p = M^d \pmod{p}$$

$$S_q = M^d \pmod{q}$$

Then these two results are combined to produce the signature S .

How does the attack work?

Suppose an attacker causes a fault in one calculation for example;

$$S'_p \neq S_p$$

but S_q is correct

The final faulty signature S' will be correct modulo q but wrong modulo

Therefore, $s' \neq m^d \pmod{p}$

but $s' \equiv m^d \pmod{q}$

The attacker can calculate,

$$g = \gcd(s'^e - m, n)$$

Because the faulty signature is correct for one prime but incorrect for the other, the GCD reveals the factor, $g = p$ or q

once p and q are known, the attacker can calculate the private key, countermeasures:

1) verify the signature before releasing it

2) Perform an additional RSA verification

3) Use fault detection and redundant calculations.

4) protect the hardware against fault injection.

40. Modular Inverse using EEA and Fermat's Little Theorem.

Ans: to the Question No-40

Find : $3^{-1} \pmod{7} = ?$

We need x such that,

$$3x \equiv 1 \pmod{7}$$

Method 1: Extended Euclidean Method,

start with $7 = 2(3) + 1$

$$\text{Therefore } 1 = 7 - 2(3)$$

So, the coefficient of 3 is -2 .

$$3(-2) \equiv 1 \pmod{7}$$

$$\text{Thus } 3^{-1} \equiv 5 \pmod{7}$$

because $-2 \equiv 5 \pmod{7}$

Method 2: Fermat's Little Theorem

For prime p :

$$a^{p-1} \equiv 1 \pmod{p}$$

$$\text{Therefore } a^{-1} \equiv a^{p-2} \pmod{p}$$

Here $p=7, a=3$

$$3^{-1} \equiv 3^{7-2} \pmod{7}$$

$$3^{-1} \equiv 3^5 \pmod{7}$$

$$(3^5)_{\text{mod } 7} = 5$$

$$243 \text{ mod } 7 = 5$$

$$\text{Therefore } 3^{-1} \equiv 5 \pmod{7}$$

comparison: $F = 1$

Euler's Theorem	Fermat's Theorem
1) Works for any modulus if inverse exists	1) Requires prime modulus

2) Usually very efficient

2) Needs modular exponentiation

3) Good for larger numbers

3) Also useful with large prime modulo

4) Gives Bezout coefficient

4) Does not directly give coefficient

11. playfair cipher

Let Ans: to the question No 11

Let use:

1) keyword: MONARCHY

2) plaintext: INSTRUMENTS AND MAELINES

It has more than 16 letters:

step 1: create 5x5 matrix

In playfair, I and J are combined

Remove repeated letters from —

MONARCHY, then add the remaining alphabet letters.

	1	2	3	4	5
1	M	O	N	A	R
2	C	H	Y	B	D
3	E	P	G	I/J	K
4	L	F	S	T	
5	U	V	W	X	Z

step 2: Divide the plaintext into digraphs

plainText: INSTRUMENTS AND MAC

CHINIES

Divide into pairs,

IN 'ST RU ME NT SA S'ND MA' CH IN E

No repeated letter - pair occurs so
no filter $\times$ is needed

step 3: Encryption Rules:

- 1) Same row $\rightarrow$ move one step right
- 2) Same column $\rightarrow$ move one step down
- 3) Different row and column $\rightarrow$ form a rectangle and take the other corners.

Diagraph Substitution:

Pair	Encryption
IN	GA
ST	TL
RU	ZM
ME	CL
NT	HR
SA	AI
ND	YR

MA → IN

CH → DH

IN → GA

ES → FI

Therefore the ciphertext is

GATLZMCLARATYRRNDHGA

Step 4: Decryption:

For decryption the rule

- Same row → move one step left
- Same column → move one step up
- Rectangle → use the opposite corners.

For example

GA → IN

TL → ST

ZM → RU

CL → ME

So, the original plaintext is -
successfully recovered

12. Playfair Cipher: structural weakness and known-plaintext Attacks

Ans: to the Question No - 012

The playfair cipher encrypts two letters at a time (digraphs), so it is much stronger than a monoalphabetic substitution cipher, which encrypts one letter at a time.

structural weakness against a digraph frequency analysis:

1) Digraph frequencies are preserved statistically: playfair hides individual letter frequencies, but it still produces ciphertext in letter pairs. common plaintext digraphs can therefore leave statistical patterns.

2) Limited digraph space:

There are only $25 \times 25 = 625$ possible ordered pairs (with I/J combined), so, frequency analysis can still be applied to ciphertext digraphs.

3) Fixed substitution rules:

A particular plaintext digraph is always transformed according to the same playfair square. This creates structural relationships that an attacker can exploit.

4) Repeated patterns can leak information

Repeated plaintext digraphs, especially common combinations, may produce repeated or related ciphertext patterns.

Vulnerability to known plaintext attacks:

If an attacker knows some plaintext and its corresponding ciphertext, they can observe how particular digraphs are transformed. Several known plaintext-ciphertext pairs can reveal relationships between letters in the 5×5 key square. Once enough positions or relationships are determined, the attacker can reconstruct much of the Playfair square and decrypt additional messages.

$$P \oplus K = C \quad \text{since } C = P \oplus K$$

$$K = P \oplus C$$

$$P \oplus K \oplus K = P \oplus C \oplus K$$

13. Two Rounds of a Simplified Feistel cipher-

Ans: to the Question No-13

For a Feistel cipher, each round is defined as:

$$L_i = R_{i-1} \oplus F(R_{i-1}, k_i)$$

Here the round function is simplified

$$F(R, k) = R \oplus k$$

Therefore, $R_i = L_{i-1} \oplus R_{i-1} \oplus k_i$

Round 1

Symbolically: $L_1 = R_0$

$$R_1 = L_0 \oplus F(R_0, k_1)$$

Since $F(R_0, k_1) = R_0 \oplus k_1$

$$L_1 = R_0$$

$$R_1 = L_0 \oplus R_0 \oplus k_1$$

Round 2: $L_1 \oplus A A A A A A A A = 1A$

$$L_2 = R_1 \text{ } 222222 = 1A$$

$$R_2 = L_1 \oplus F(R_1, K_2)$$

$$L_2 = R_1$$

$$R_2 = L_1 \oplus R_1 \oplus K_2$$

Numerical calculation:

given:

$$L_0 = 0 \times A A A A A A A A$$

$$R_0 = 0 \times F F F F F F F F$$

$$K_1 = 0 \times 0 F F F F F F$$

$$K_2 = 0 \times 3 3 3 3 3 3 3 3$$

Round 1:

$$F(R_0, K_1) = F F F F F F F F \oplus 0 F F F F F F F$$

$$= F F F F F F F F$$

Therefore $L_1 = F F F F F F F F$

and

$$R_1 = \text{AAAAAAAAAA} \oplus \text{FFFFFFFFFF}$$

$$R_1 = \text{55555555}$$

$$\text{so, } L_1 = \text{0xFOFOFOFO}, R_1 = \text{0x55555555}$$

Round 2: $F(R_1, K_2)$

$$= \text{55555555} \oplus \text{33333333}$$

$$\text{AAAA} = \text{66666666}$$

Therefore

$$L_2 = R_1 = \text{55555555}$$

$$R_2 = L_1 \oplus \text{66666666}$$

$$= \text{F0F0F0F0} \oplus \text{66666666}$$

$$\text{70707070} \oplus \text{07070707} = R_2 = \text{96969696}$$

14. Avalanche effect

Ans. to the Question No - 14

Formal definition:

The avalanche effect means that a small change in the input of a cryptographic algorithm should produce a large and unpredictable change in the output.

Ideally, changing one input bit should change approximately 50% of the output bits.

For an n bit output

Expected changed bits $\approx \frac{n}{2}$

For an 8 bit cipher:

$$\frac{8}{2} = 4$$

So, changing one input bit should ideally change about 4 output bits.

Toy 8 bit example:
consider the following toy substitution

Input	Output
00000000 00000001	10110010 01001101
Now flip only the last input bit	
00000000 00000001	10110010 01001101

compare
Expected changed bits
01001101
01001101
11111111

All 8 output bit changed

Therefore, changed bit = $8/8 = 100\%$
which is greater than the required
half 50%

Why is avalanche effect important?

- 1) It hides the relationship between ciphertext and plaintext.
- 2) A tiny plaintext change should not produce a predictable ciphertext change.
- 3) It makes statistical and differential attacks more difficult.
- 4) It provides confusion and diffusion important properties of secure block cipher.

15: Brute Force Resistance: Password vs PIN

Ans: to the Question No-15

Given GPU rate:

10^9 guesses/second

A. 8-character alphanumeric password alphabet contains: 62 characters.

Therefore total possibilities,

$$62^8 = 218,340,105,584,896$$

So, approximately:

$$= 21834 \times 10^4$$

Worst case time:

Worst case means the attacker tries every possibility.

$$T = \frac{62^8}{10^9} = 218,340.1 \text{ S}$$

convert to hours:

$$\frac{2183404}{3600} = 606.5 \text{ hours}$$

$$\approx 2.53 \text{ days}$$

Expected case time:

on average, the correct password is found halfway through the search:

$$T_{\text{avg}} = \frac{628}{2 \times 10^9} = 169,170.05 \text{ s}$$

$$\approx 30.3 \text{ hours} \approx 1.26 \text{ days}$$

3.6 digit - PIN

each digit has 10 possibilities

$$10^6 = 1,000,000$$

so there are 1,000,000 possibilities

Worst case:

$$\frac{1,000,000}{2} = 500,000$$

$$= 1 \text{ millisecond}$$

Expected case time

$$\text{Time} = \frac{1000000}{2 \times 10^9}$$

$$= 0.0005 \text{ s}$$

$$= 0.5 \text{ millisecond}$$

limi

Implications for password policy:

1) Longer passwords are much harder to brute force.

2) System should use rate limiting, account lockout, MFA and other protections against repeated guessing.

3) A 6 digit PIN has only 1 million combinations.

4) Password should preferably be long and randomly generated rather than relying only on short

complexity requirement of IV.

3) The calculation assumes an ~~unrestricted~~ unrestricted offline attacker at exactly 10^9 guesses/s.

16. Comparison of ECB, CBC, CFB, OFB and CTR.

Ans: to the Question No - 016

1. ECB - electronic codebook

Diagram:

$P_1 \rightarrow [E] \rightarrow C_1$

$P_2 \rightarrow [E] \rightarrow C_2$

$P_3 \rightarrow [E] \rightarrow C_3$

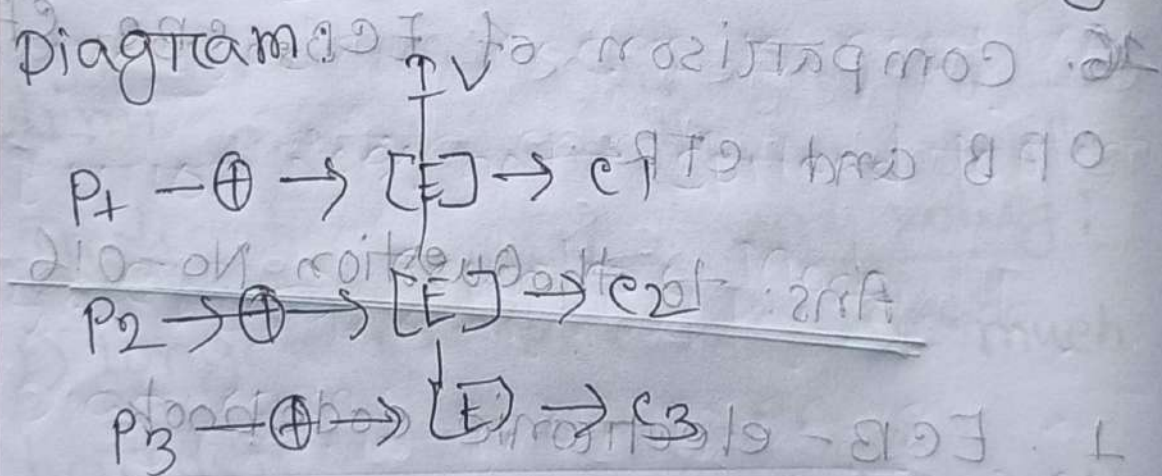
- Parallel encryption: Yes
- Parallel decryption: Yes
- Bit-flip error: Affects only the corresponding block.

- IV/Nonce: Not required

- Problem: Same plaintext gives the same ciphertext. So patterns are visible.

2. CBC — Cipher Block chaining

Diagram



Parallel encryption: No

Parallel decryption: Yes

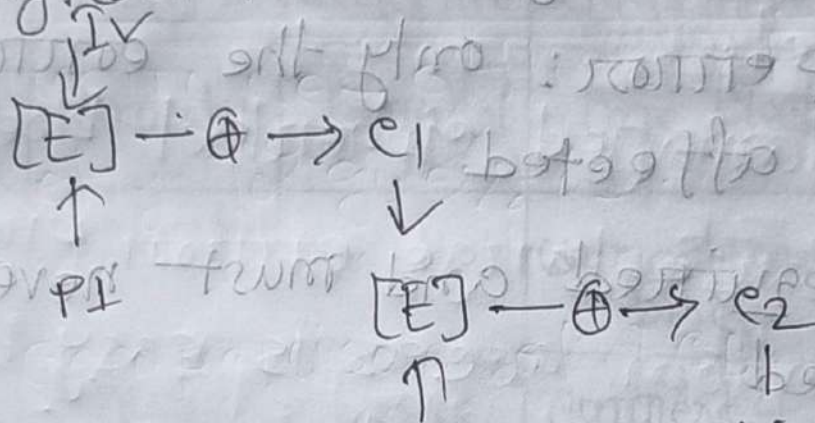
Bit-flip error: Corrupts the current block and the next block.

IV: Required and should be unpredictable

Encryption is sequential because each block depends on the previous ciphertext.

3. CFB - Cipher Feedback

Diagram:



Parallel encryption: No

Parallel decryption: Yes

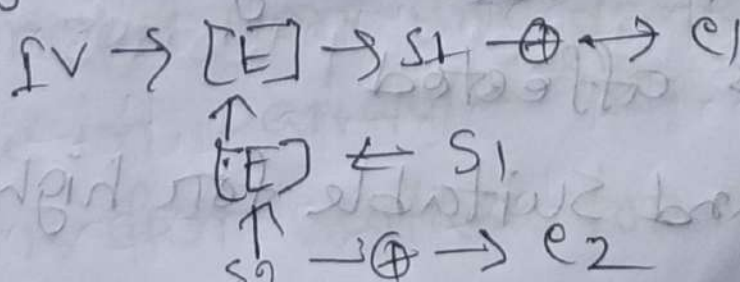
Bit-flip error: Affects the current and next block/segment

IV: Required / and must be unique.

It behaves like a stream mode

4. OFB - Output Feedback

Diagram:



Parallel encryption: No
Parallel decryption: No
Bit-flip error: only the corresponding bit is affected
IV: Required and must never be reused

Error does not spread to other block.

5. CTR $\rightarrow$ Counter Mode
Diagram

Nonce + 1 $\rightarrow [E] \rightarrow S_1 \oplus P_1 \rightarrow C_1$

Nonce + 2 $\rightarrow [E] \rightarrow S_2 \oplus P_2 \rightarrow C_2$

Nonce + 3 $\rightarrow [E] \rightarrow S_3 \oplus P_3 \rightarrow C_3$

Parallel encryption: yes

Parallel decryption: yes

Bit-flip error: Only the corresponding bit is affected

Fast and suitable for high speed data

Comparison

Mode	Encryption	Decryption	Error Propagation	IV/None
ECB	Parallel	Parallel	One block	No
CBC	Sequential	Parallel	Current + next block	Yes
CFB	Sequential	Parallel	Current + next block	Yes
OFB	Sequential	Sequential	One bit	Yes, never reuse
CTR	Parallel	Parallel	One bit	Yes, Never reuse

a) Live video stream over a lossy network

Reasons:

- 1) A bit error affects only the corresponding bit
- 2) Error does not spread to many blocks
- 3) Encryption and decryption can be done in parallel
- 4) It is fast suitable for continuous data.

(b) Static disk image

- 1) Disk data can be processed in parallel.
- 2) A damaged ciphertext block does not cause large error propagation.
- 3) Random access is easy because CTR uses a counter.

18. AES-128 Rijndael Key Schedule

Ans. to the Question No - 18

Defination

The AES-128 key schedule generates 10 round keys from the original 128 bit key. Each round key has 128 bits (16 bytes).

For AES-128:

128 bit key = 4 words

Let the words be, W_0, W_1, W_2, W_3

The main steps are:

Rotword $\rightarrow$ subword $\rightarrow$ XOR with Round $\rightarrow$ XOR

given 128 bit key

Take:

$K_0 = 000102030405060708090A0B$
 $0C0D0E0F$

So,

$W_0 = 00\ 01\ 02\ 03$

$W_1 = 04\ 05\ 06\ 07$

$W_2 = 08\ 09\ 0A\ 0B$

$W_3 = 0C\ 0D\ 0E\ 0F$

Round 1: Find K_1

Step 1: Rotword

Take W_3 :

$W_3 = 0C\ 0D\ 0E\ 0F$

Rotate left by one byte:

Rotword = $0D\ 0E\ 0F\ 0C$

Step 2: Subword

Apply the AES S box:

0D $\rightarrow$ D7

0E $\rightarrow$ AB

0F $\rightarrow$ 76

0C $\rightarrow$ FE

Therefore: subword = D7 AB 76 FE

Step 3: Rcon

For Round 1: 01 00 00 00 = R1

Rcon = 01 00 00 00

XOR: 0D AB 76 FE 00 = W1

D7 AB 76 FE 00 = W2

$\oplus$ 01 00 00 00

D6 AB 76 FE 00

step 4: Generate new words

$$\begin{aligned} W_4 &= W_2 \oplus D6A3B76FE \\ &= 00010203 \oplus D6A3B76FE \\ &= D6AA74FD \end{aligned}$$

$$\begin{aligned} W_5 &= W_1 \oplus W_4 \\ &= 04050607 \oplus D6AA74FD \\ &= D2AF72FA \end{aligned}$$

$$\begin{aligned} W_6 &= W_2 \oplus W_5 \\ &= 08090A0B \oplus D2AF72FA \end{aligned}$$

$$\begin{aligned} W_7 &= W_3 \oplus W_6 \\ &= 0C0D0E0F \oplus DAA678F4 \\ &= D6AB76FE \end{aligned}$$

Therefore, key k_1

$$\begin{aligned} &= D6AA74FDD2AF72FADAA67 \\ &\quad 8F1D6AB76FE \end{aligned}$$

Round 2: find key

$$W_7 = D6 \oplus AB \oplus 76 \oplus FE \oplus 02 = 76$$

Step 1: Rot word = AB 76 FE D6

Step 2: Subword

$$AB \rightarrow 62$$

$$76 \rightarrow 38$$

$$FE \rightarrow BB$$

$$D6 \rightarrow F6$$

So, Subword = 62 38 BB F6

Step 3: Reon

For round 2: $2W \oplus 2W = FW$

$$\text{Reon2} = 02 \ 00 \ 00 \ 00$$

$$\begin{array}{cccc} 62 & 38 & BB & F6 \\ & 02 & 00 & 00 \end{array}$$

$\oplus$

$$\hline 60 \ 38 \ BB \ F6$$

F2A A44 75 F7A 507 7AF A42D

772 F8A 21A 78

step 4: Generates new words.

$$W_8 = W_4 \oplus 6038BBFF$$

$$= D6AA74FD \oplus 6038BBFF$$

$$= B692CF0B$$

$$W_9 = W_5 \oplus W_8$$

$$= D2AF2FA \oplus B692CF0B$$

$$= 643DBDFF$$

$$W_{10} = W_6 \oplus W_9$$

$$= DAA678F1 \oplus 643DBDFF$$

$$= BE9BCE50$$

$$W_{11} = W_7 \oplus W_{10}$$

$$= D6AB76FE \oplus BE9BCE50$$

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 20 & 50 & 10 & 10 \\ 50 & 10 & 10 & 60 \end{bmatrix}$$

19. AES-Mix columns

Ans. to the Question No - 19

Mix columns is a transformation used in AES. It works on one column (4 bytes) of the AES state at a time. Each column is multiplied by a fixed matrix in the finite field $GF(2^8)$.

The AES irreducible polynomial is

$$x^8 + x^4 + x^3 + x + 1$$

In hexadecimal, this polynomial is represented by 0x11B.

→ Mix columns matrix

The fixed matrix is

$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 04 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix}$$

Take one input column.

DB
13
55
45

Then:

DB
13
53
45

02 03 01 01
01 02 03 01
01 01 02 03
03 01 01 02

8E
4D
Ad
Be

2A or F(28) Arithmetic

In AES: $03 \times a = (02 \times a) \oplus a$

20. AES Side-channel Attack Resistance

Ans: to the Question No - 20

AES is a cryptographically strong, but its implementation can leak secret information through side channels such as execution time and power consumption.

1) Timing / Cache Timing Attack:

In a table based AES, implementation the S-box is stored in a lookup table.

Secret key $\rightarrow$ S-box $\rightarrow$ Memory $\rightarrow$ execution time

Different S-box position may be different CPU Cache lines. An attacker can measure the execution time and use this information to guess secret key bytes.

counter measure: constant Time S box;
use a constant-time S box implementation where memory access and execution time do not depend on secret data.

advantages: Reduces timing and cache attacks.

Disadvantages: May require more CPU and slower.

2. Differential Power Analysis (DPA)

DPA studies the power consumption of a device while AES is running.

AES operation $\rightarrow$ power consumption

many measurements $\rightarrow$ key information

Small changes in power consumption can reveal information about intermediate AES values and eventually the secret key.

masking:

$$X' = X \oplus R$$

where R is a Random mask
advan: much harder

Disadvantage: Require extra
calculation.

21. Derive RSA key Generation, encryption and
decryption using Euler's theorem, and prove that
decryption gives the original message, even when
 $\gcd(M, n) \neq 1$.

Answer to the question no: 21

1. key Generation

choose two prime number:

p, q

calculate:

$$n = pq$$

Euler's totient :

$$\phi(n) = (p-1)(q-1)$$

choose a public exponent e such that :

$$1 < e < \phi(n), \gcd(e, \phi(n)) = 1$$

choose d such that :

$$ed \equiv 1 \pmod{\phi(n)}$$

So : * Public key : (e, n)

* private key : (d, n)

2. Encryption

for message M : $C = M^e \pmod{n}$

3. Decryption

The receiver calculates :

$$M = C^d \pmod{n}$$

Therefore, $M = (M^e)^d = M^{ed} \pmod{n}$

Since

$$ed \equiv 1 \pmod{\phi(n)}$$

we can write: $ed = 1 + k\phi(n)$

If $\gcd(M, n) = 1$ by Euler's theorem:

$$M^{\phi(n)} \equiv 1 \pmod{n}$$

Therefore, $M^{ed} = M^{1+k\phi(n)}$

$$= M (M^{\phi(n)})^k$$

$$\equiv M (1)^k$$

$$M^{ed} \equiv M \pmod{n}$$

When $\gcd(M, n) \neq 1$

Euler's theorem cannot be directly used.

We prove it separately modulo p and q .

Since $n = pq$, consider modulo p .

* If $p \nmid M$, Euler/Fermat gives:

$$M^{p-1} \equiv 1 \pmod{p}$$

* If $p \mid M$, then: $M^{ed} \equiv 0 \equiv M \pmod{p}$

The same is true mod q .

Therefore, $m^d \equiv M \pmod{p}$

and $m^d \equiv M \pmod{q}$

Since p and q are different primes, by Chinese
Remainder Theorem (CRT):

$$m^d \equiv M \pmod{pq}$$

Since $n = pq$, $(m^d)^d \equiv M^d \pmod{n}$

So RSA decryption correctly recovers the original
message for all M , including $\gcd(M, n) \neq 1$.

$$(E^{-1}, E)$$

22. For $p = 11$ and $q = 13$, generate RSA keys, sign hash $H(m) = 23$ and verify the signature (q, bon) $M = \text{bon}$

Ans: to the question No - 22

Step 1: Find n

$$n = pq = 11 \times 13 = 143$$

Step 2: Find $\phi(n)$

$$\phi(n) = (11-1)(13-1)$$

$$= 10 \times 12 = 120$$

Step 3: choose e

choose: $e = 7$

because $\gcd(7, 120) = 1$

so public key is $(7, 143)$

Step 4: Find d

We need

$$ed \equiv 1 \pmod{120}$$

$$7d \equiv 1 \pmod{120}$$

$$\text{Since } 7 \times 103 = 721 = 6(120) + 1$$

$$\text{Therefore } d = 103$$

private key $(103, 143)$

step 5: sign the hash

$$\text{given: } H(M) = 23$$

signature:

$$s = H(M)^d \pmod{n}$$

$$s = 23^{103} \pmod{143}$$

Now show modular exponentiation:

$$23^2 = 529 \equiv 100 \pmod{143}$$

$$23^4 = 100^2 = 10000 \equiv 53 \pmod{143}$$

$$23^8 = 53^2 = 2809 \equiv 87 \pmod{143}$$

$$23^{16} = 87^2 = 7569 \equiv 133 \pmod{143}$$

$$23^{32} = 133^2 \equiv 100 \pmod{143}$$

$$23^{64} = 100^2 \equiv 53 \pmod{143}$$

since $103 = 64 + 32 + 4 + 2 + 1$

Therefore, $S = 6$

$$23^{103} = 23^{64} 23^{32} 23^4 23^2 23^1$$

$$\equiv 53 \times 100 \times 53 \times 100 \times 23 \pmod{143}$$

step by step $S = (M)^H$

$$53 \times 100 \equiv 87$$

$$87 \times 53 \equiv 133$$

$$133 \times 100 \equiv 100$$

$$100 \times 23 \equiv 87$$

$$\text{So, } S = 87$$

ver

step 6: verify signature

Receiver calculator

$$S^e \pmod{n}$$

$$87^7 \pmod{143}$$

modular steps:

$$87^2 = 7569 = 133 \pmod{143}$$

$$87^4 \equiv 133^2 \equiv 100 \pmod{143}$$

Therefore,

$$87^7 \equiv 87^4 + 87^2 \times 87$$

$$\equiv 100 \times 133 \times 87$$

$$100 \times 133 \equiv 87$$

$$87 \times 87 \equiv 23 \pmod{143}$$

$$\text{Thus } 87^7 \pmod{143} \equiv 23$$

$$H(M) = 23$$

Therefore signature is valid

23. Derive the ESP Tunnel mode overhead percentage formula and calculate it for a 1200 byte payload, 8 byte header, 16 byte authentication data and 2 byte padding

Ans: to the question No - 23

Let, p = payload size

H = ESP header size

A = authentication data size

D = padding size

Total ESP overhead:

$$O = H + A + D$$

So, overhead percentage compared with the original payload is:

$$\text{Overhead \%} = \frac{H + A + D}{p} \times 100$$

$$P = 1200 \text{ bytes}$$

$$H = 8 \text{ bytes}$$

$$A = 16 \text{ bytes}$$

$$D = 2 \text{ bytes}$$

$$O = 8 + 16 + 2 = 26 \text{ bytes}$$

$$\text{overhead \%} = \frac{26}{1200} \times 100$$

$$= 2.17\%$$

$$\text{total transmitted size} = 1200 + 26$$

$$= 1226 \text{ bytes}$$

$$\text{overhead} = \frac{1}{\text{payload size}}$$

$$\boxed{CH = 04 = 2 + 2 \times 1 = 4}$$

24. Perform DH key exchange for $p = 23$, $g = 5$, $a = 6$, $b = 15$. Show the shared secret and explain MITM attack and its solution.

Ans to the Q: No - 24

$$p = 23, g = 5$$

Aleyna's private key $a = 6$

Ans 6

Bob's private key $b = 15$

Step 1: calculate public key

$$\text{Aleyna } A = g^a \mod p$$

$$A = 5^6 \mod 23$$

$$5^2 = 25 \equiv 2$$

$$5^4 \equiv 2^2 = 4$$

$$5^6 = 5^4 \times 5^2 \times 5$$

$$= 4 \times 2 \times 5 = 40 \equiv 17$$

So, Aleya sends $A = 17$

Badal:

$$B = g^b \pmod{p}$$

$$B = 5^{15} \pmod{23}$$

using powers,

$$5^2 = 2, \quad 5^4 = 4, \quad 5^8 = 16$$

$$5^{15} = 5^8 \cdot 5^4 \cdot 5^2 \cdot 5$$

$$= 16 \times 4 \times 2 \times 5$$

$$= 19 \pmod{23}$$

So, Badal sends $B = 19$

Step 2: calculate shared secret

$$\text{Aleya: } k = B^a \pmod{p}$$

$$k = 19^6 \pmod{23}$$

$$19^2 = 361 \equiv 16$$

$$19^4 \equiv 16^2 \equiv 256 \equiv 3$$

$$19^6 = 19^4 \times 19^2$$

$$= 3 \times 16 = 48 \equiv 2$$

$$\text{Badal: } k = a^b \text{ mod } p$$

$$k = 17^{15} \text{ mod } 23$$

$$17^2 = 13$$

$$17^4 = 8$$

$$17^8 = 18$$

$$17^{15} = 17^8 17^4 17^2 17$$

$$= 18 \times 8 \times 13 \times 17$$

$$= 2 \text{ (mod } 23)$$

$$\text{Therefore, } k_{\text{actual}} = k_{\text{Badal}} = 2$$

Q25. Compare RSA key and DH key

Feature	RSA Key Transport	DH/ECDH
Main operation	RSA encryption/decryption	modular/exponentiation operation
Example security	3072-bit RSA	3072-bit DH or 256-bit ECDH

Computation	RSA private key operation can be expensive	ECDH is generally efficient
key agreement	NO	Yes
Forward secrecy	NO	Yes

Why (E)DHE is preferred?

RSA key transport:

The server's long term private RSA key is used to recover the session key. If an attacker records old encrypted traffic and later steals the RSA private key, old sessions may be decrypted.

(E)DHE: uses temporary (private) key for each session. After the session, those temporary keys are discarded.

30. Compare RSA, ElGamal and EEC key sizes and explain why EEC is useful for IoT devices.

Ans. to the question - 30

System	~128-bit security
RSA	3072 bit
ElGamal	3072 bit
EEC	256 bit

Why EEC is preferred:

- 1) Smaller key size
- 2) Less memory and storage needed
- 3) Faster computation
- 4) Uses less power.
- 5) Good for smart cards and IoT sensors with limited resources

31. Define preimage, second-pre image and collision resistance. Derive birthday attack complexity.

Ans to the question No-31

For an n bit hash

1) preimage resistance: given hash h , it is difficult to find m such that

$$H(m) = h$$

$$\text{cost} \rightarrow 2^n$$

2) second preimage resistance:

given m_1 , it is difficult to find a different m_2 such that

$$H(m_1) = H(m_2)$$

$$\text{cost} \rightarrow 2^n$$

3) Collision resistance

It is difficult to find any two different message M_1, M_2 such that $H(M_1) = H(M_2)$.

Birthday Attack:

After about k random hashes, collision probability becomes significant when:

$$\frac{k^2}{2^n} \approx 1$$

$$\text{So, } k^2 \approx 2^n$$

$$k \approx 2^{n/2}$$

Therefore, an n -bit hash has approximately $2^{n/2}$ collision security.

$$\text{For 128 bit} \rightarrow \frac{128}{2} = 64$$

$$\text{For 256 bit} \rightarrow \frac{256}{2} = 128$$

So, modern system prefer 256 bit hashers for about 128 bit collision security.

32. Explain Merkle Damg and construction, length, attack and how HMAC/SHA-3 prevent it

Ans. to the Question No - 32

Message

$M_1 \rightarrow [\text{compression}] \rightarrow H_1$

$M_2 \rightarrow [\text{compression}] \rightarrow H_2$

$M_3 \rightarrow [\text{compression}] \rightarrow H_3$

Hash

Each block uses the previous chaining values as a input.

length Extension Attack
If an attacker knows,

$H(M)$
and the length of M , they can calculate a valid hash for,

$M || \text{padding} || 2$
without the knowledge of the original

33. Explain the main stages of the SHA-256 compression function with a diagram.

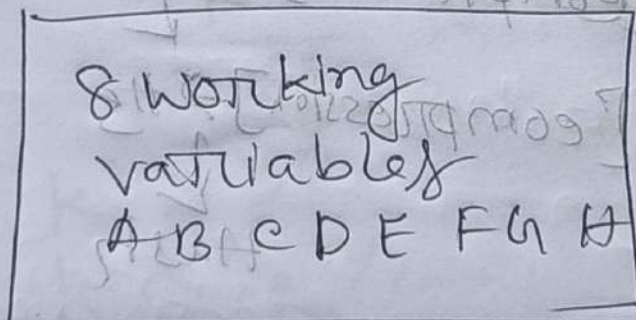
Ans: to the Question No-33

Message Block (512 bits)



Message schedule

$W_0, W_1, \dots, W_{63}$



Round Function
(64 round)

Updated A-H values

New Hash/chain variables

34. SVP and SIVP -

Ans. to the Question No - 34

Shortest Vector Problem (SVP):

given a lattice L , find a non-zero - shortest vector,

$$v \in L, v \neq 0$$

$$\text{such that } \|v\| = \lambda_1(L) = \min_{u \in L, u \neq 0} \|u\|.$$

Shortest Independent Vector Problem

given an n -dimensional lattice L , find n linearly independent lattice vectors $v_1, \dots, v_n$ such that the maximum length is minimized,

$$\max(\|v_i\|) = \lambda_n(L)$$

Why hard for quantum computers?

- 1) SVP and SIVP are computationally hard lattice problems.
- 2) No efficient classical algorithm is known for solving them in general.
- 3) No efficient quantum algorithm for SVP/SIVP. Shor efficiently solves factoring and discrete logarithm.

35. CRYSTALS - kyber and Module LWE

Ans. to the Question No - 35

core idea:

kyber is based on the Module Learning with Errors (Module LWE) Problem.

A simplified LWE equation is,

$$b = As + e$$

where, A = public Random matrix

s = Secret vector

e = small random noise/ error

b = public value.

The noise e makes it difficult to recover s from A and b .

kyber KEM:

- 1) key generation: creates a public key and secret key using Module LWE.
- 2) Encapsulation: sender uses the public key and randomness to create a ciphertext and shared secret.
- 3) Decapsulation: receiver uses the secret key to recover the same shared secret.

why secure:

1) Recovering the secret from noisy lattice equations is believed to be hard.

2) They also have no known efficient quantum solution like Shor's algorithm.

3) Hence Kyber is designed to provide post quantum security.

36. Kerberos Authentication Protocol.

Ans: to the question No-36

1) As exchange:

client sends ID c and request a ticket for the TGS.

2) As Reply:

As sends;

- ke, tgs = session key between client and TGS.

- TGT = Ticket Granting Ticket encrypted using the TGS's secret key

The client's password is used to protect the response, so the password itself is not sent over the network.

③ TGS Request:

client sends the TGT+ Authenticator to TGS to request access to servers

④ TGS Reply:

TGS sends;

1) k_{es} = session between client and server

2) Tickets = ticket for server.

⑤ client-server exchange

⑥ server Reply

37. Plain Hash Mac vs HMAC

Ans: to the Question No-37

Plain Hash Mac.

$$MAc = H(k || M)$$

1) In a plain hash Mac, the secret key k is placed before the message M .

2) Many hash functions use a Merkle-Damgaard structure.

3) An attacker knowing $H(k || M)$ may continue hashing extra data M' .

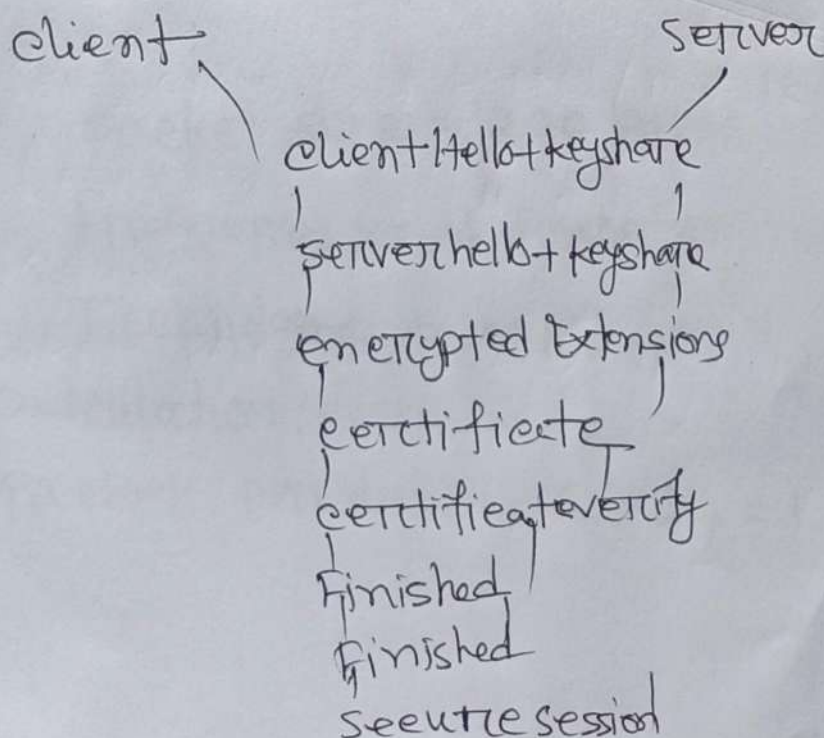
HMAC solution:

$$H((K \oplus \text{opad}) \parallel H((K \oplus \text{ipad}) \parallel M))$$

- 1) HMAC first performs an inner hash and then an outer hash.
- 2) The secret-derived values $K \oplus \text{ipad}$ and $K \oplus \text{opad}$ protect both stages.
- 3) The attacker can not directly extend the final HMAC value.
- 4) Therefore, HMAC eliminates the length extension vulnerability.

38. TLS 1.3 Handshake

Diagram:



key Derivation:

client key share

+

server key share

+

(E) DHE shared secret

+

HKDF

+

Handshake keys

+

Application keys

39. lot Bandwidth & Duty cycle
Budget

Ans. to the question No-39

given :

packet size = 1200 bytes

Frequency = 1 packet/minute

Technology = LoRaWAN

calculation:

packet per day = $60 \times 24 = 1440$ packet/day

Data per day:

$$1200 \times 1440 = 1728,000 \text{ bytes/day}$$

$$\text{In KB} \rightarrow \frac{1728000}{1024} = 1687.5 \text{ KB/day}$$

Data Cycle:

At 1% duty cycle means maximum channel time of $60 \times 60 \times 0.01 = 36 \text{ s/h}$

40. Aescon vs AES-128 for IoT.

Feature	Aescon	AES - 128
Design	Lightweight cryptography	General-purpose block cipher
Memory	generally small footprint	can require more resource
energy	Designed for low energy devices	efficient especially with hardware acceleration
Security	Modern AEAD design	very well established